

Storm Kristin, Portugal, January 2026

Storm Kristin made landfall on the Iberian Peninsula in late January 2026, bringing significant precipitation to Portugal and Spain. This report summarizes station-level precipitation and temperature data from NOAA's [Global Historical Climatology Network – Daily \(GHCN-D\)](#) for stations within a bounding box covering the Iberian Peninsula and surrounding region (35–45°N, 15°W–5°E).

Station Metadata

Station metadata is sourced from NOAA's `ghcnd-stations.txt`, pre-processed by `scripts/read_ghcn_daily.py` and stored locally as `stations_output.csv`. Stations are filtered to those with `.dly` observation files on disk and falling within the bounding box.

Station Metadata (first 10 rows)

275 stations in bounding box

Code	Country	State	Station ID	Station Name	Lat	Lon	Elev (m)
AG	Algeria		AG000060390	ALGER-DAR EL BEIDA	36.72	3.25	24
AG	Algeria		AGE00135039	ORAN-HOPITAL MILITAIRE	35.73	0.65	50
AG	Algeria		AGE00147705	ALGIERS-VILLE/UNIVERSITE	36.78	3.07	59
AG	Algeria		AGE00147706	ALGIERS-BOUZAREAH	36.80	3.03	344
AG	Algeria		AGE00147707	ALGIERS-CAP CAXINE	36.80	3.04	38
AG	Algeria		AGE00147708	TIZI OUZOU	36.72	4.05	222
AG	Algeria		AGE00147709	FORT NATIONAL	36.63	4.20	942
AG	Algeria		AGE00147712	ORLEANSVILLE (CHLEF)	36.17	1.34	112
AG	Algeria		AGE00147714	ORAN-CAP FALCON	35.77	0.80	78
AG	Algeria		AGE00147716	NEMOURS (GHAZAOUET)	35.10	−1.85	83

Observations

Observations are loaded from GHCN-D `.dly` files — one file per station. Each file contains daily measurements for all recorded elements (precipitation, max/min temperature, snow

depth, etc.) across the full station history. Quality-flagged records are removed and values are converted to metric units (precipitation in mm, temperature in °C).

10 / 275 stations (9s) 20 / 275 stations (26s) 30 / 275 stations (34s) 40 / 275 stations

Observations (first 10 rows)

Long format — one row per station × date × element

Station ID	Station	Country	Date	Element	Value	M Flag	Q Flag	S Flag
AG000060390	ALGER-DAR EL BEIDA	Algeria	1940-01-01	PRCP	0.0			E
AG000060390	ALGER-DAR EL BEIDA	Algeria	1940-01-01	TMAX	22.4			G
AG000060390	ALGER-DAR EL BEIDA	Algeria	1940-01-01	TMIN	4.7			E
AG000060390	ALGER-DAR EL BEIDA	Algeria	1940-01-02	PRCP	0.0			E
AG000060390	ALGER-DAR EL BEIDA	Algeria	1940-01-02	TMAX	20.2			G
AG000060390	ALGER-DAR EL BEIDA	Algeria	1940-01-02	TMIN	8.8			E
AG000060390	ALGER-DAR EL BEIDA	Algeria	1940-01-03	PRCP	1.8			E
AG000060390	ALGER-DAR EL BEIDA	Algeria	1940-01-03	TMAX	21.0			G
AG000060390	ALGER-DAR EL BEIDA	Algeria	1940-01-03	TMIN	11.0			E
AG000060390	ALGER-DAR EL BEIDA	Algeria	1940-01-04	PRCP	18.5			E

Precipitation

Daily precipitation (PRCP) is pivoted to wide format: one row per station, one column per calendar date. Only stations reporting on 0% of days in the analysis window are retained.

Precipitation — Wide Format (first 10 rows)

Values in mm; minimum completeness threshold: 0%

Station ID	Station	Country	Completeness	2026-01-25	2026-01-26	2026-01-27
FR000007510	BORDEAUX-MERIGNAC	France	100%	21.2	3.0	17.6
FR000007630	TOULOUSE-BLAGNAC	France	100%	2.6	0.0	12.0
FR000007747	PERPIGNAN	France	100%	2.4	0.0	4.0
FRE00104116	TARBES - OSSUN	France	100%	29.3	1.4	8.3
FRE00104907	MONTÉLIMAR	France	100%	0.6	0.0	42.2
FRE00106200	MILLAU	France	100%	5.0	0.0	9.7
FRE00106203	MONT-DE-MARSAN	France	100%	17.6	4.4	21.5
FRE00106205	ST-GIRONS	France	100%	18.2	3.8	10.6
FRE00106207	MONTPELLIER-AEROPORT	France	100%	1.0	0.0	12.7
FRM00007535	GOURDON	France	100%	5.4	0.0	4.2

Temperature

Max temperature (TMAX) is pivoted using the same approach, with the same 0% completeness threshold.

Max Temperature — Wide Format (first 10 rows)

Values in °C; minimum completeness threshold: 0%

Station ID	Station	Country	Completeness	2026-01-25	2026-01-26	2026-01-27
AGM00060452	ARZEW	Algeria	86%	16	20	20
FR000007510	BORDEAUX-MERIGNAC	France	100%	12	12	10
FR000007630	TOULOUSE-BLAGNAC	France	100%	10	13	11
FR000007747	PERPIGNAN	France	100%	13	14	10
FRE00104116	TARBES - OSSUN	France	100%	14	11	10
FRE00104907	MONTELMAR	France	100%	8	10	9
FRE00106200	MILLAU	France	100%	5	7	8
FRE00106203	MONT-DE-MARSAN	France	100%	10	12	9
FRE00106205	ST-GIRONS	France	100%	11	11	11
FRE00106207	MONTPELLIER-AEROPORT	France	100%	12	14	11

Data Preparation

Precipitation totals are summed across the analysis window and joined with station coordinates and country metadata for mapping and analysis.

Precipitation Map Data (first 10 rows)

Total and daily precipitation (mm), Jan 25–Jan 31, 2026

Station ID	Station	Country	Total (mm)	2026-01-25	2026-01-26	2026-01-27
FR000007510	BORDEAUX-MERIGNAC	France	70.9	21.2	3.0	17.6
FR000007630	TOULOUSE-BLAGNAC	France	29.6	2.6	0.0	12.0
FR000007747	PERPIGNAN	France	9.8	2.4	0.0	4.0
FRE00104116	TARBES - OSSUN	France	86.5	29.3	1.4	8.3
FRE00104907	MONTELMAR	France	43.6	0.6	0.0	42.2
FRE00106200	MILLAU	France	29.7	5.0	0.0	9.7
FRE00106203	MONT-DE-MARSAN	France	88.4	17.6	4.4	21.5
FRE00106205	ST-GIRONS	France	58.8	18.2	3.8	10.6
FRE00106207	MONTPELLIER-AEROPORT	France	17.5	1.0	0.0	12.7
FRM00007535	GOURDON	France	24.8	5.4	0.0	4.2

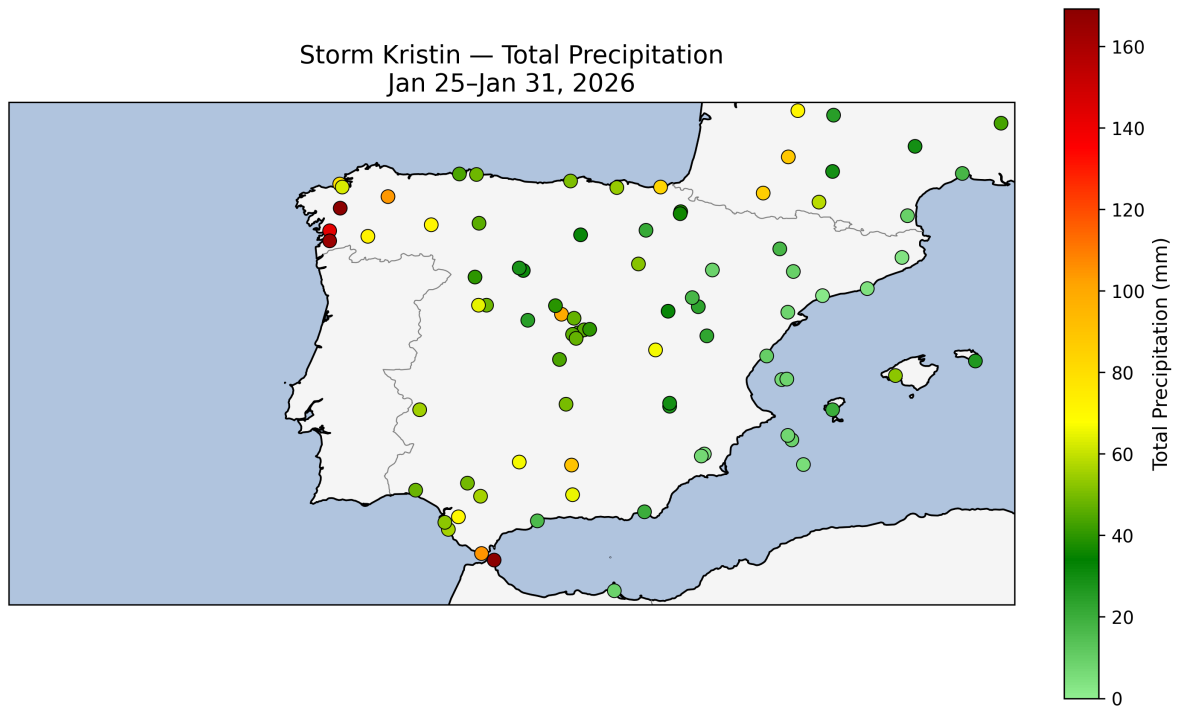
Summary Statistics

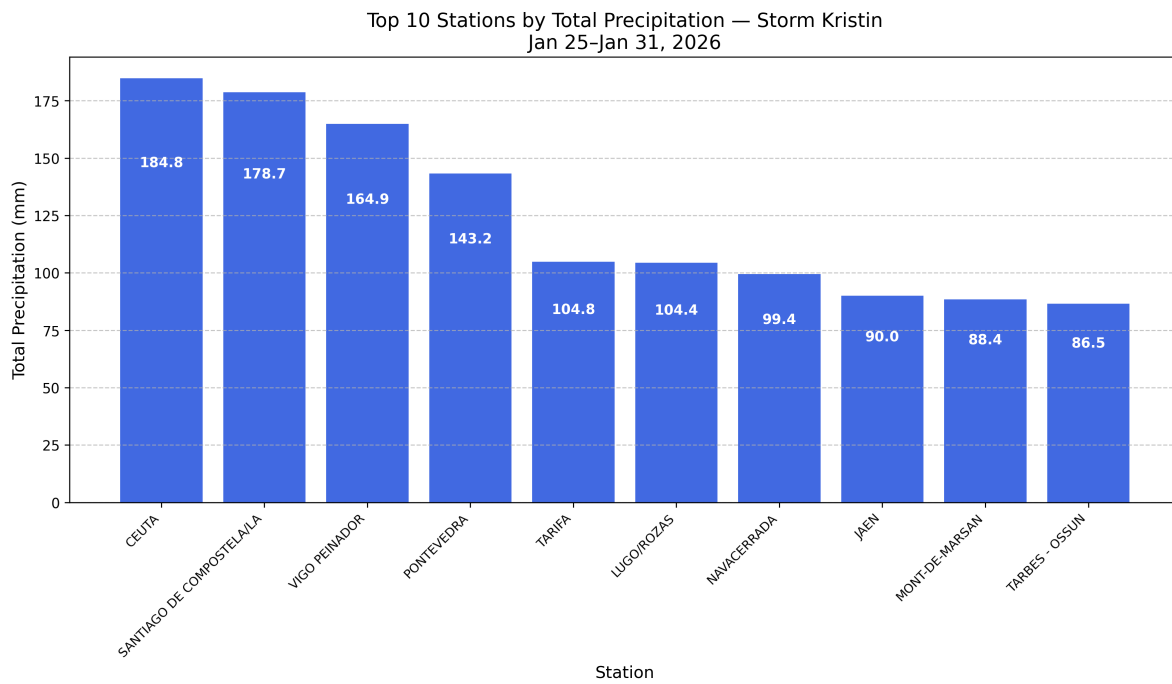
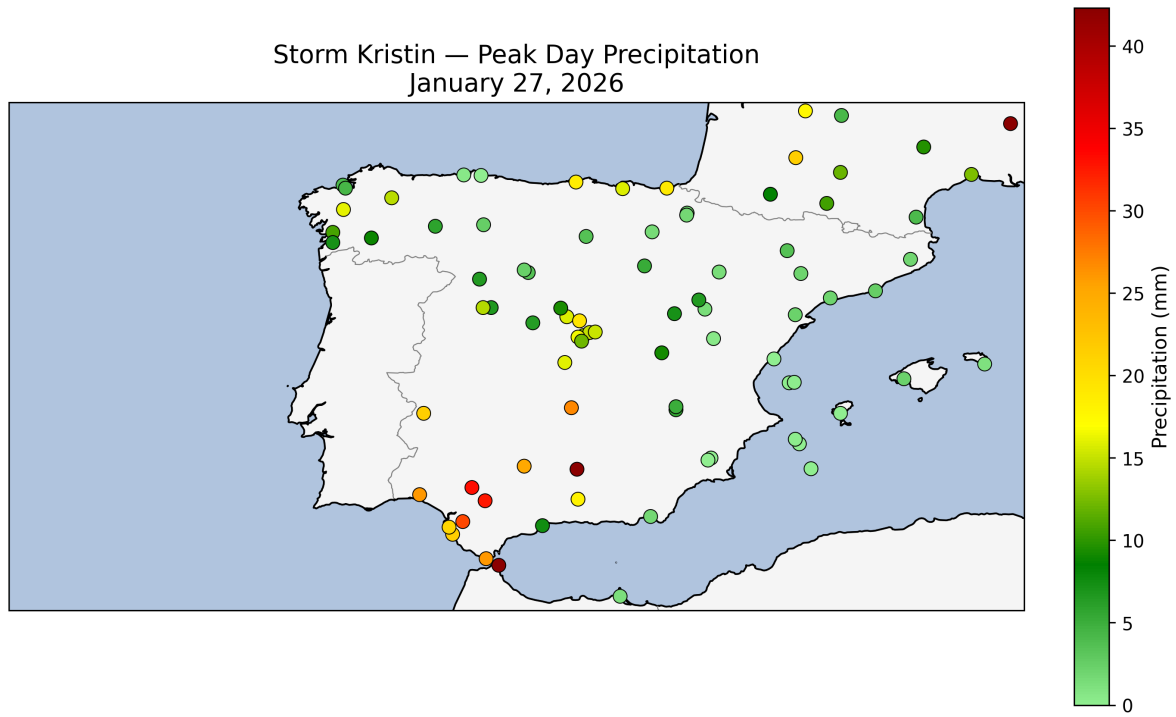
Stations in Bounding Box by Country		
Country	Code	Stations
Spain	SP	198
Algeria	AG	33
France	FR	20
Portugal	PO	18
Morocco	MO	5
Gibraltar [United Kingdom]	GI	1

Total Precipitation by Country					
Jan 25–Jan 31, 2026					
Country	Code	Stations	Median (mm)	Min (mm)	Max (mm)
Spain	SP	75	43.7	2.5	184.8
France	FR	10	36.7	9.8	88.4

Daily Precipitation Statistics — All Reporting Stations			
Date	Mean (mm)	Median (mm)	Max (mm)
2026-01-25	5.1	2.0	30.8
2026-01-26	9.5	4.4	92.2
2026-01-27	10.8	6.7	67.8
2026-01-28	9.3	7.7	53.6
2026-01-29	5.0	5.0	26.0
2026-01-30	5.5	4.1	26.0
2026-01-31	1.8	0.2	24.1

Visualizations





Storm Kristin — Daily Mean Precipitation Across Reporting Stations
Portugal Region, 2026

